Renal Physiology:

Glomerular Filtration Rate, Renal Blood Flow and Clearance

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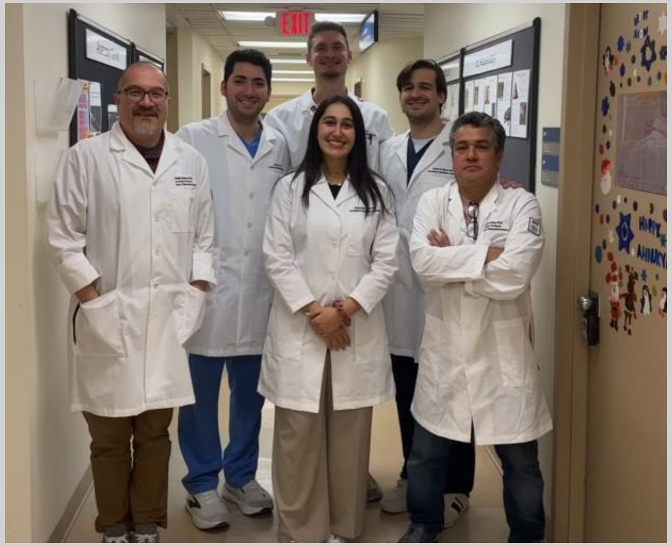


About your Scholar....

Case Study

A 58-year-old man presents to his primary care physician for a routine follow-up visit. He has a 15-year history of type 2 diabetes mellitus and hypertension. His medications include metformin and amlodipine, although he admits to inconsistent adherence. He reports increasing fatigue and swelling in his ankles over the past several months. Physical examination reveals blood pressure of 158/96 mm Hg and bilateral pitting edema of the lower extremities.

Laboratory studies demonstrate:

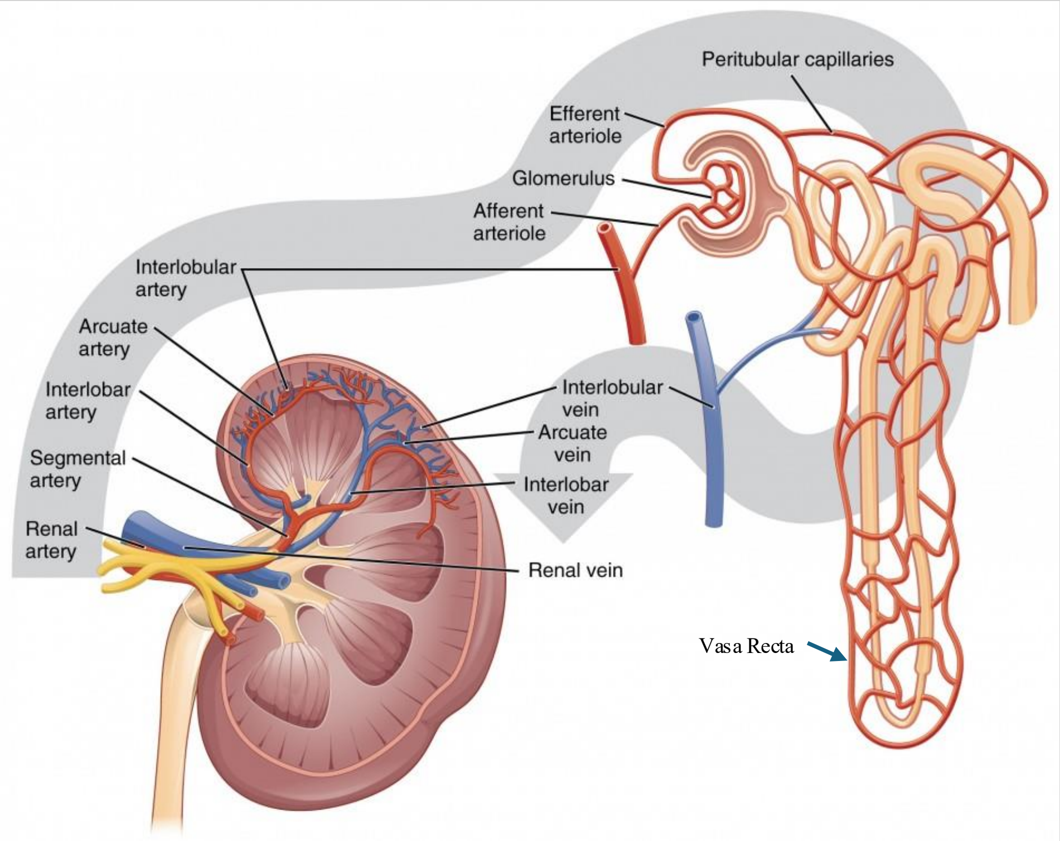
Tests	Results	Normal Values
Fasting serum glucose	**212 mg/dL**	70-99 mg/dL
HbA1c	**9.4%.**	<5.7% mg/dL
Serum creatinine	**1.8 mg/dL**	0.7-1.3 mg/dL
BUN	**32 mg/dL**	8-24 mg/dL
Urine albumin	**Positive 78 mg/g**	<30 mg/g
Urinalysis	**Proteinuria without hematuria**	pale yellow, clear

- What is your differential? Why?
- Throughout the disease course, what physiological changes occurred within this patient's renal system? Would a renal biopsy show anything abnormal?
- How are GFR, oncotic & hydrostatic pressures, filtration ability, and autoregulation of blood flow implicated?
- What could have prevented this from occurring? How can you help this patient?

Learning Objectives

1. Demonstrate understanding of the anatomy of the renal system, especially in relation to blood supply, filtration, the renal cortex, medulla, tubular system, and glomerulus.
2. Correlate gross anatomy and function of the kidneys with GFR, resistance, blood pressure, hydrostatic and oncotic pressures.
3. Determine how autoregulation and vasoactive influences affect the kidneys' role in homeostasis.
4. Formulate enough knowledge in filtration and blood flow of the renal system to apply to a pathological process case study.
5. Apply what you have learned to board-style multiple-choice questions.

Gross Renal Blood Flow and Anatomy



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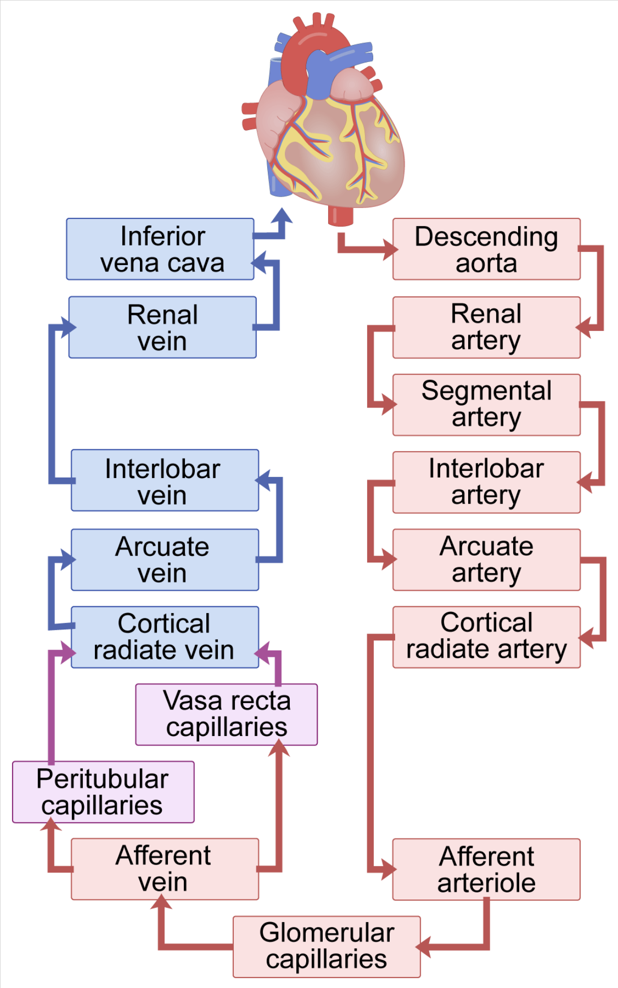
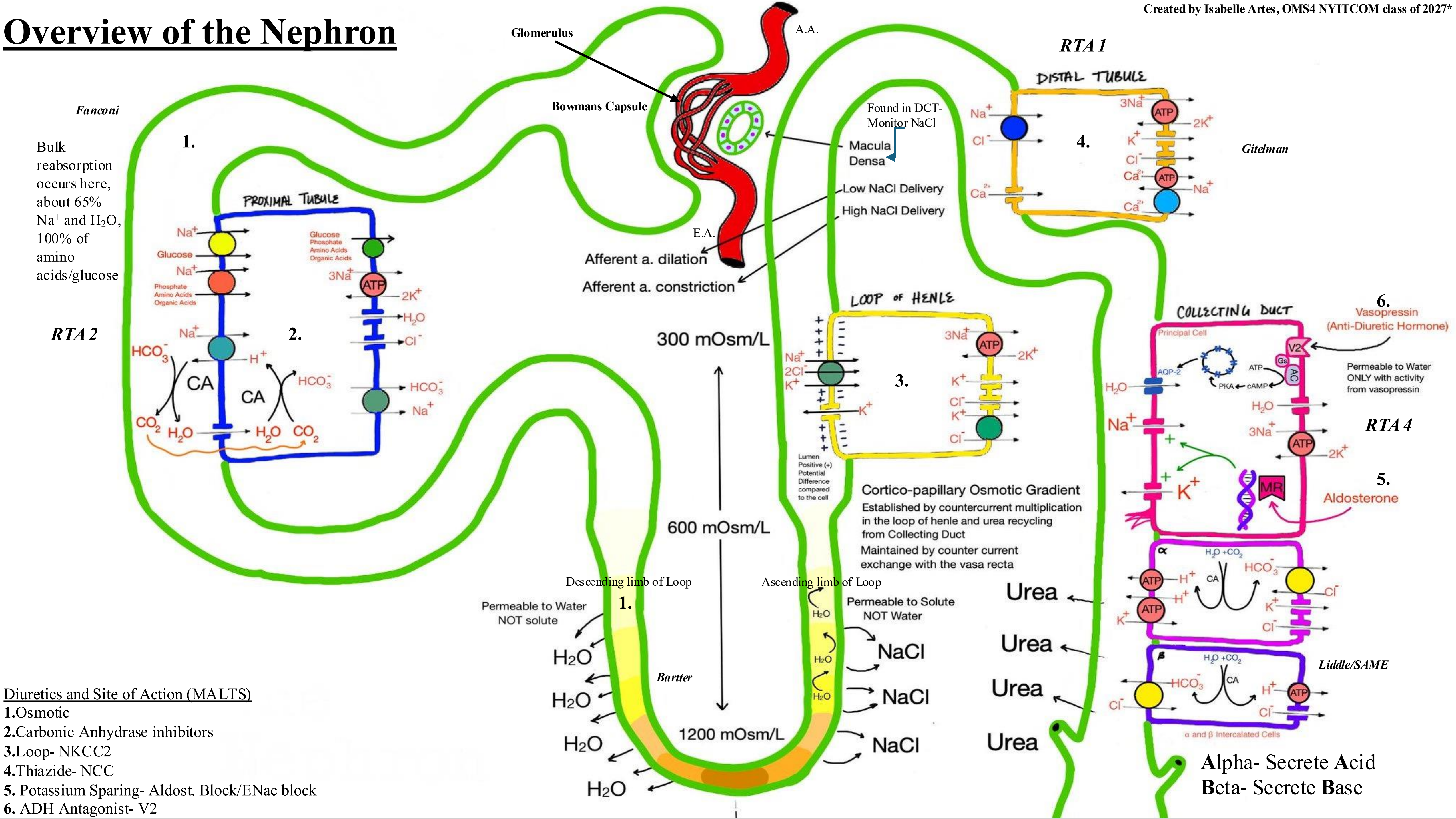


Human Bio Media. (n.d.). *Kidney: Blood supply*. https://www.humanbiomedia.org/kidney-blood-supply-lesson/

- Blood flow through the Kidneys (RBF) is massive considering the small size of the **retroperitoneal** organ (about 0.5% of body weight)
- About 1L/min or 20% of resting Cardiac Output (HR*SV) reaches the kidneys!
- Each kidney has a total volume of 150cc's, meaning that **over 3X** volume flows through each kidney every minute
- **Left vs Right Renal Vein**- Left is longer, more prone to compression (**Nutcracker syndrome**) and has the left gonadal vein exiting it (increased risk for **Varicocele** on left). Left kidney is also preferred for **transplant**.
- Most blood stays in the **Cortex** (PCT and DCT) after passing through glomeruli by way of the **Peritubular Capillaries**. 5-10% of blood goes to the "salty" **Medulla** (Loop of Henle) by way of the **Vasa Recta**. Why have these two separate paths? Aide in concentrating urine by keeping solute concentration high in the medulla by countercurrent exchange of water and to supply the medulla with O2/nutrients (discussed further in other lectures). **Medulla is extremely sensitive to Hypoxia and ischemic damage (ATN)**

Overview of the Nephron

Created by Isabelle Artes, OMS4 NYITCOM class of 2027*

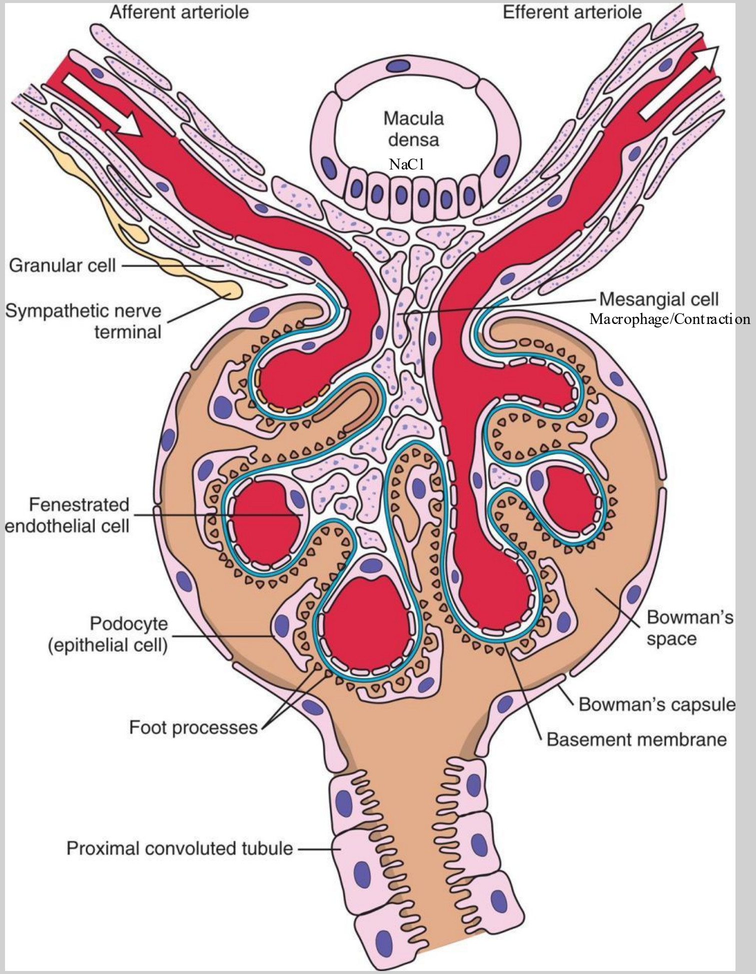


Diuretics and Site of Action (MALTS)

1. Osmotic
2. Carbonic Anhydrase inhibitors
3. Loop- NKCC2
4. Thiazide- NCC
5. Potassium Sparing- Aldost. Block/ENac block
6. ADH Antagonist- V2

A Closer Look at the Glomerulus and Bowman's Capsule (Renal Corpuscle)

Juxtaglomerular cells = Granular Cells
(mechanoreceptive cells that store and release Renin)



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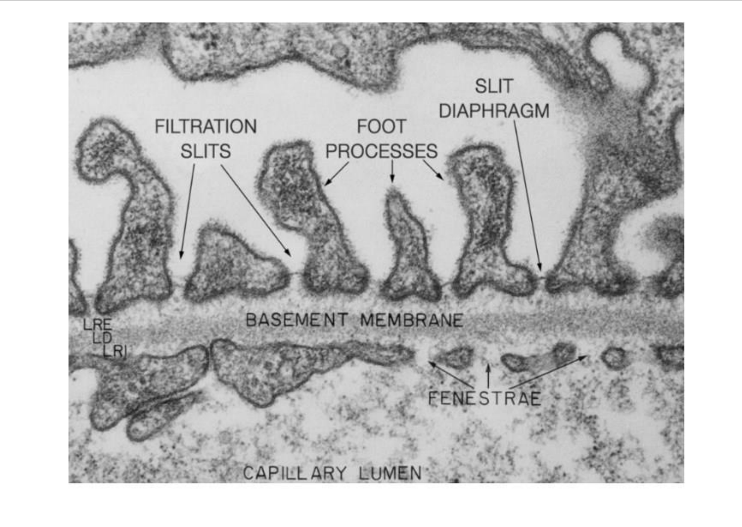

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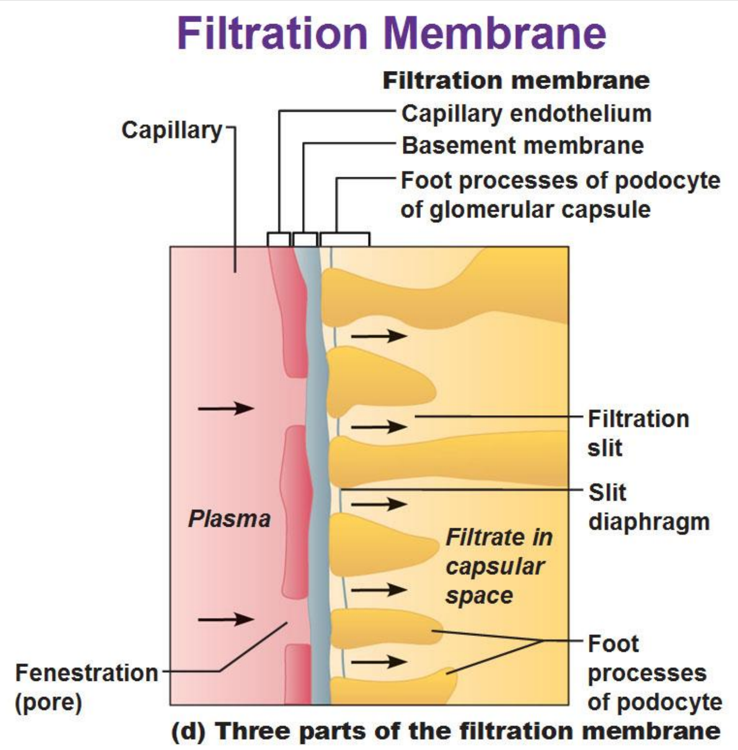

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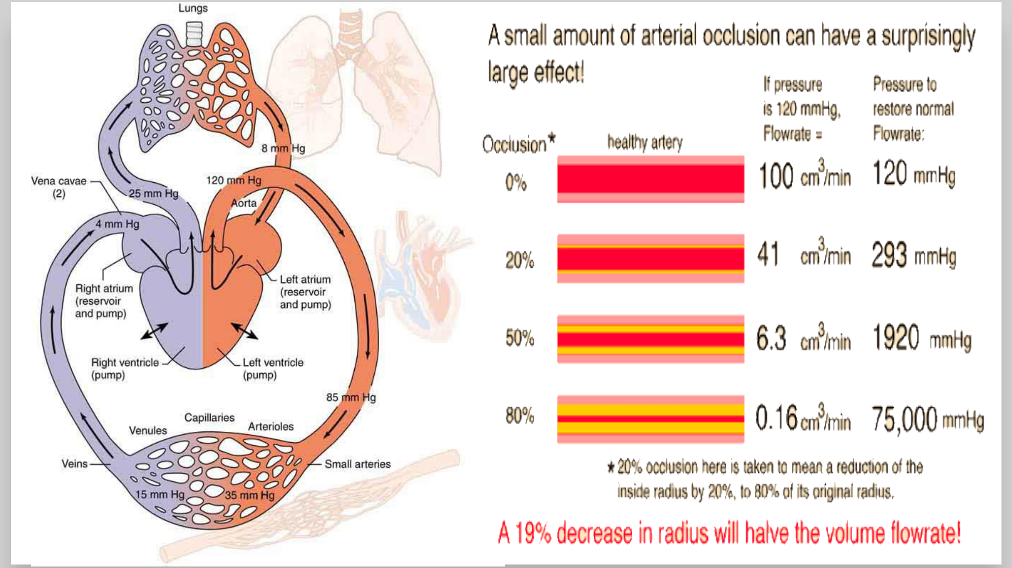
- Approximately 1 million Glomeruli are in each kidney and decrease in numbers as we age.
- Instead of Arteriole-Venule, it is **Arteriole-Arteriole (Afferent-Capillary-Efferent)**

- In patients undergoing nephrectomy, it is expected that a proportional 50% decrease in renal function will occur post-procedure
- Hypertrophy of remaining kidney can occur, increasing to greater then 50% of original two kidney renal function, with GFR nearly doubling in some cases (due to decrease a.a. resistance)

OpenStax. (n.d.). *Viscosity and laminar flow; Poiseuille's law*. In *College Physics*. BCcampus OpenEd. https://pressbooks.bccampus.ca/collegephysics/chapter/viscosity-and-laminar-flow-poiseuilles-law/

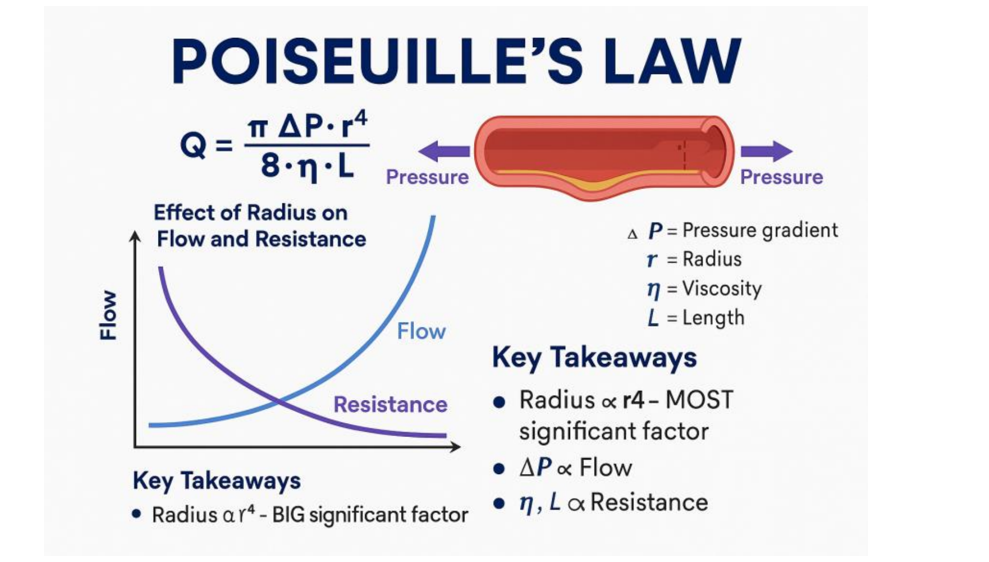
Renal Blood Flow, Resistance and Blood Pressure



Blood Flow/**Q** (mL/min)=△**P/R**= Difference of pressure across vasculature (inflow-outflow)/Resistance

Resistance/**R**=**8*L****n***/pi*r^4**= 8*Length of vessel*viscosity of blood/pi* radius of the vessel to the fourth power

- Renal blood flow follows **Ohm's law**: flow = (ΔP) / resistance, with resistance determined by **Poiseuille's law**.
- A key determinant of resistance is vessel radius, which is proportional to the **fourth power—making it the most powerful regulator of blood flow.**
- Changes in afferent and efferent arteriole diameter therefore have a major impact on renal blood flow, blood pressure, and glomerular filtration rate.



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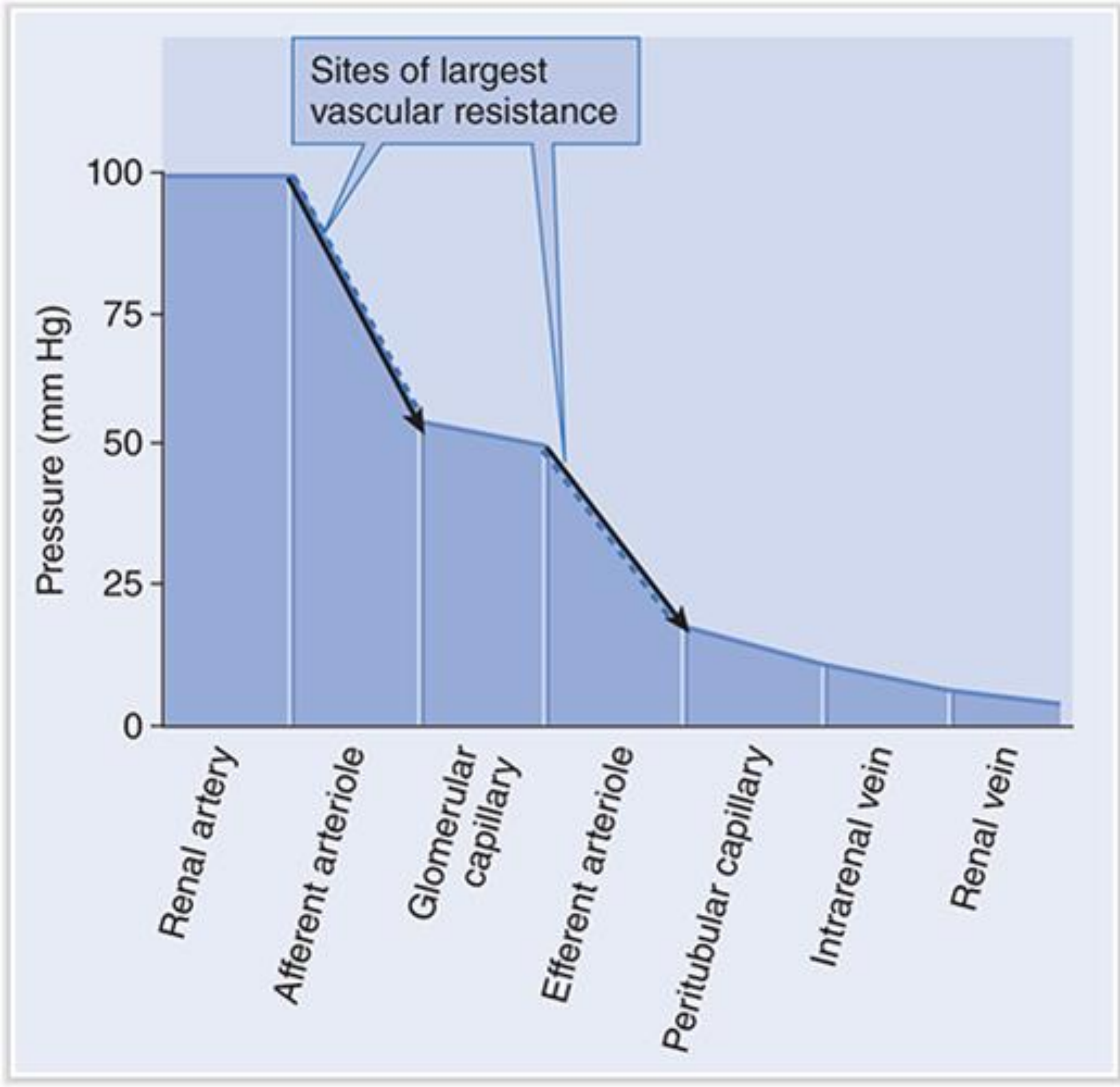
Resistance Change as Blood Travels into smaller Renal Vessels

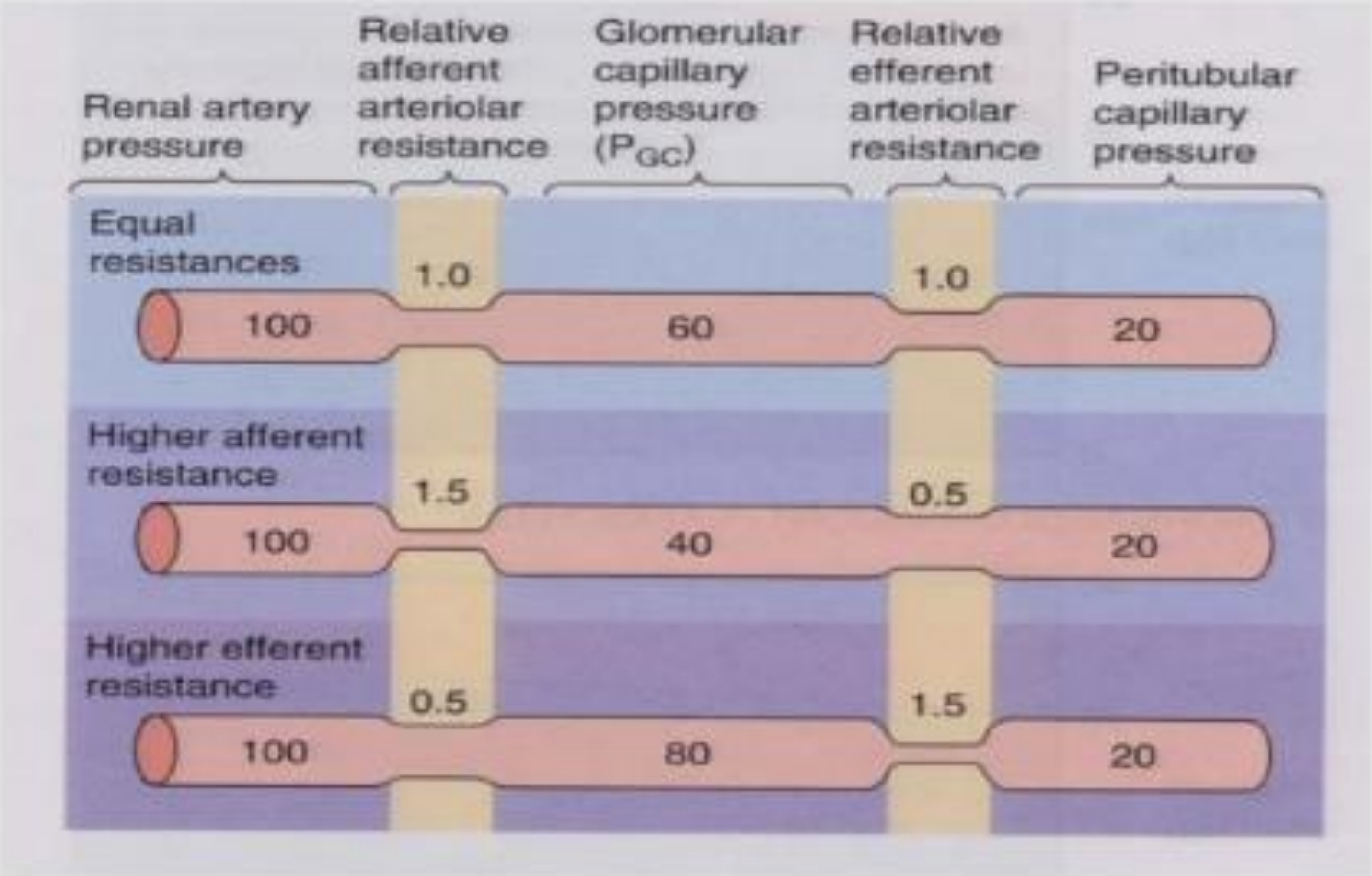
- **Net Filtration** in Afferent Arterioles and Glomerular capillary's, **Net reabsorption** in peritubular capillary's
- As blood flows through more resistance (smaller "r"/radius) pressure will **decrease**
- The location of the glomerular capillaries, between the sites of high resistance, results in their having a much higher pressure than the peritubular capillaries.
- **Selective constriction or relaxation of afferent and efferent arterioles allows for sensitive control of hydrostatic pressure (Blood pressure in mm Hg)** in intervening glomerular capillary and thus glomerular filtration



Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

From "Medical Physiology second edition", Boron and Boulpaep

Changes in Afferent and Efferent Resistances



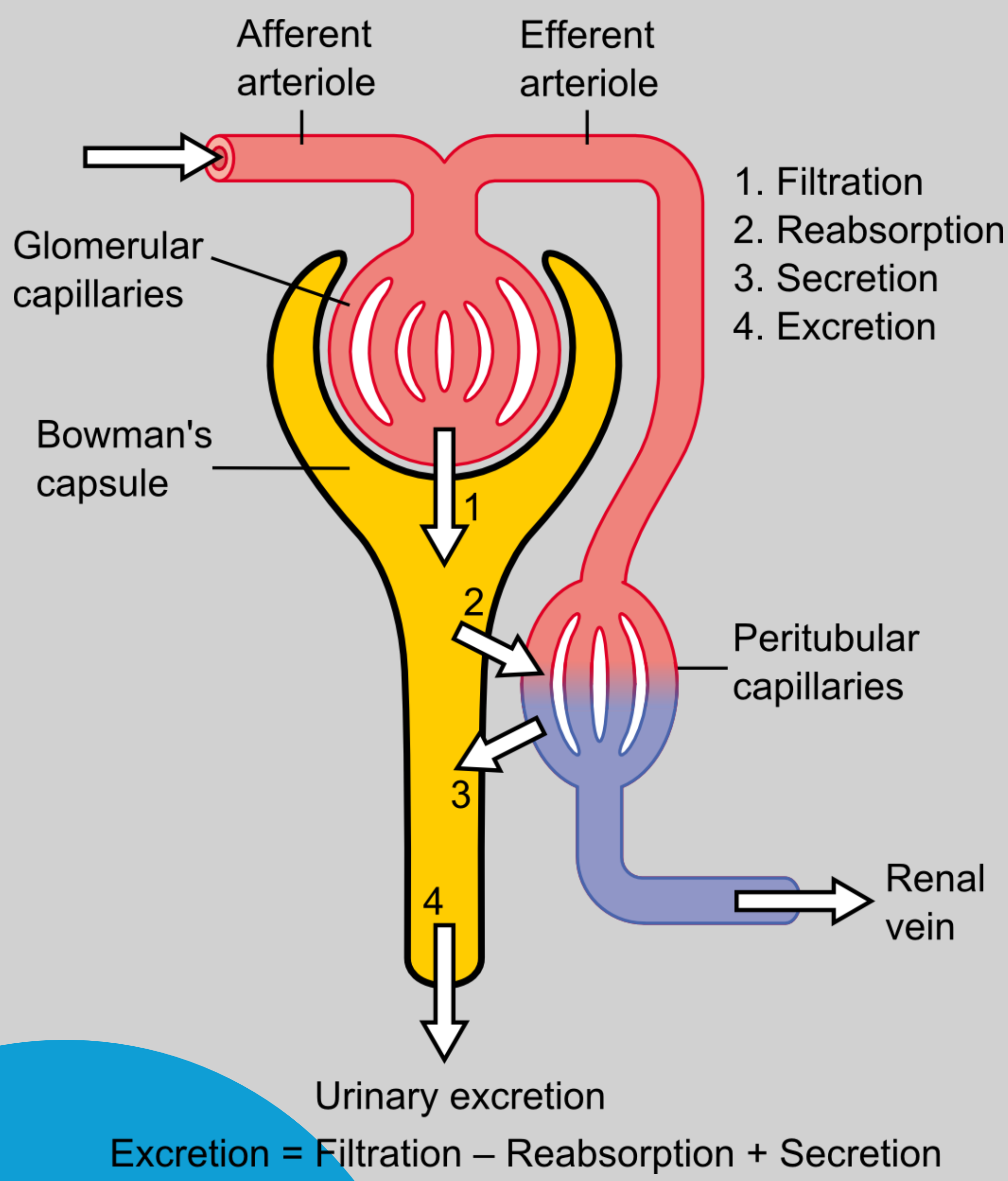
Medical Physiology Boron and B

Feature	**Renal Blood Flow**	**Renal Plasma Flow**
Definition	Total blood volume passing through kidneys each minute	Volume of cell free plasma passing through the kidneys each minute
Normal value	About 1L/min (About 20% of CO)	600-650 ml/min
Function	Overall oxygen and nutrient delivery to the kidneys	Actual filterable fluid the kidneys will get per minute
Formula	**RBF** = RPF/(1-Hematocrit)	**RPF** = RBF*(1-Hematocrit) (HCT) **eRPF** = V*Upah/Ppah=Clearance of PAH

- **Q vs total RBF (difference):**
- **Q:** flow within individual renal vessels (local hemodynamics)
- **RBF:** total blood volume delivered to kidney (global flow)
- **Hematocrit (Hct):**
- % of blood = RBCs
- Normal:
  - Men: 41–50%
  - Women: 36–47%
- ↑ Hct → polycythemia/dehydration
- ↓ Hct → anemia
- 1 − Hct ≈ plasma fraction
- ↑ Hct → ↑ viscosity (η) → ↑ resistance → ↓ renal blood flow
- **PAH (para-aminohippuric acid):**
- Used to estimate **effective renal plasma flow (eRPF)**
- Almost completely cleared from plasma in one pass through kidney
- Clearance of PAH ≈ RPF
- Alternatives: radioactive tracers (renal scintigraphy)
- **PAH clearance equation variables:**
- Upah : urine PAH concentration
- V: urine flow rate (mL/min)
- Ppah : plasma PAH concentration

<u>Glomerular Filtration</u>

- How much blood is filtered through the kidneys daily?
- How much urine is produced?
- How much reabsorption of solutes and water take place (substances that are filtered initially but return to the blood stream)?



Wikipedia contributors. (n.d.). *Ultrafiltration (kidney)*. In *Wikipedia*. Retrieved June 11, 2026, from https://en.wikipedia.org/wiki/Ultrafiltration_(kidney)

Terminology

- Things in tubule that stay in tubule (only about 1% of filtrate) end up in **urine**
- **Ultrafiltration/Filtration**- occurs at the renal corpuscle, high pressure environment force arterial contents into bowman's capsule
- **Glomerular Filtrate**- fluid left after ultrafiltration occurs
- **Reabsorption**- Initially filtered into the tubule, but eventually brought back into the blood stream
- **Secretion**- the transfer of waste products into the nephron to eventually stay in the nephron and eventually urine
- **Excretion**- Urination out of the body
- **Clearance**- rate at which a substance is removed from the blood by the kidney and put into urine to be excreted(U*V/P [urine]*urine flow rate/[plasma])
- **Glomerular Filtration Rate (GFR)**- is the volume of fluid filtered from the blood by the kidneys' glomeruli into the kidney tubules per unit of time. It is the best overall indicator of kidney function and reflects how effectively the kidneys are filtering waste products and excess fluid from the blood. Expressed as mL/min, usually a **24 hour Urine Test**
- **eGFR – Blood test** for creatinine level estimates used routinely in medicine, based on formulas (see slide 19 for links)

What factors influence the Ability to filter?

- ***What is in the blood that will reach the renal corpuscle?***
  - Small, Positively Charged inorganic molecules have similar concentration in filtrate to the blood (Electrolytes, Glucose, Amino acids, Urea, Creatinine, Vitamins) and filter freely
  - Large negatively charged proteins and blood cells do not filter freely in physiological conditions
- ***What barriers at the filtration site exist?***
  - 3 Layered Glomerular Filtration Barrier (in order of inside the glomerulus to the outside 1-3)
    - **Fenestrated Endothelial Cells** of the Glomerular Capillaries- Size Exclusion to less than 100 nM
    - **Capillary Basement Membrane (GBM)-** Type 4 collagen, heparan sulfate, glycoproteins and proteoglycans, NEGATIVELY CHARGED, keep out large negative molecules and proteins
    - **Visceral epithelial** layer with **Podocyte Foot Processes-** Size Exclusion to less then 50 nM, foot processes with Slit Diaphragms
  - **K_f**= Product of membrane permeability to water and surface area of filter barrier, can be different in other tissues, under active control
- ***What pressures in bowman space and the glomerulus influence filtration?***

Net filtration pressure: $[(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$

- P_{GC} = **hydrostatic pressure in glomerular capillary (GC)**
  - Blood pressure in GC: 60 mmHg
  - Favors filtration
- P_{BS} = **hydrostatic pressure in Bowman's space (BS)**
  - Fluid pressure in BS: ~15 mmHG
  - Opposes filtration
- π_{GC} = **oncotic pressure in the glomerular capillary**
  - Blood proteins (Albumin!), endothelial glycocalyx, basement membrane
  - Opposes filtration
- π_{BS} = **oncotic pressure of the filtrate in Bowman's space**
  - Very little protein in BS (x → 0); omitted from equation
  - Favors filtration (but not calculated...)

- Measured in **mL/min**

P_{GC} =hydrostatic pressure of glomerular capillary

π_{GC} =colloid osmotic pressure of glomerular capillary

$$GFR = K_f[(P_{GC} - P_{BS}) - (\pi_{GC} - \pi_{BS})]$$

K_f =glomerular permeability

P_{BS} =hydrostatic pressure of fluid in Bowman's space

π_{GC} = colloid osmotic pressure of fluid in Bowman's space

Substance	Molecular Weight (Da)	Effective Molecular Radius* (nm)	Filtrate (UF_x/P_x)
Na^+	23	0.10	1.0
K^+	39	0.14	1.0
Cl^-	35	0.18	1.0
H_2O	18	0.15	1.0
Urea	60	0.16	1.0
Glucose	180	0.33	1.0
Sucrose	342	0.44	1.0
Polyethylene glycol	1,000	0.70	1.0
Inulin	5,200	1.48	0.98
Lysozyme	14,600	1.90	0.8
Myoglobin	16,900	1.88	0.75
Lactoglobulin	36,000	2.16	0.4
Egg albumin	43,500	2.80	0.22
Bence Jones protein	44,000	2.77	0.09
Hemoglobin	68,000	3.25	0.03
Serum albumin	69,000	3.55	<0.01

*The effective molecular radius is the Einstein-Stokes radius, which is the radius of a sphere that diffuses at the same rate as the substance under study.
Data from Pitts RF: Physiology of the Kidney and Body Fluids, 3rd ed. Chicago: Year Book, 1974.

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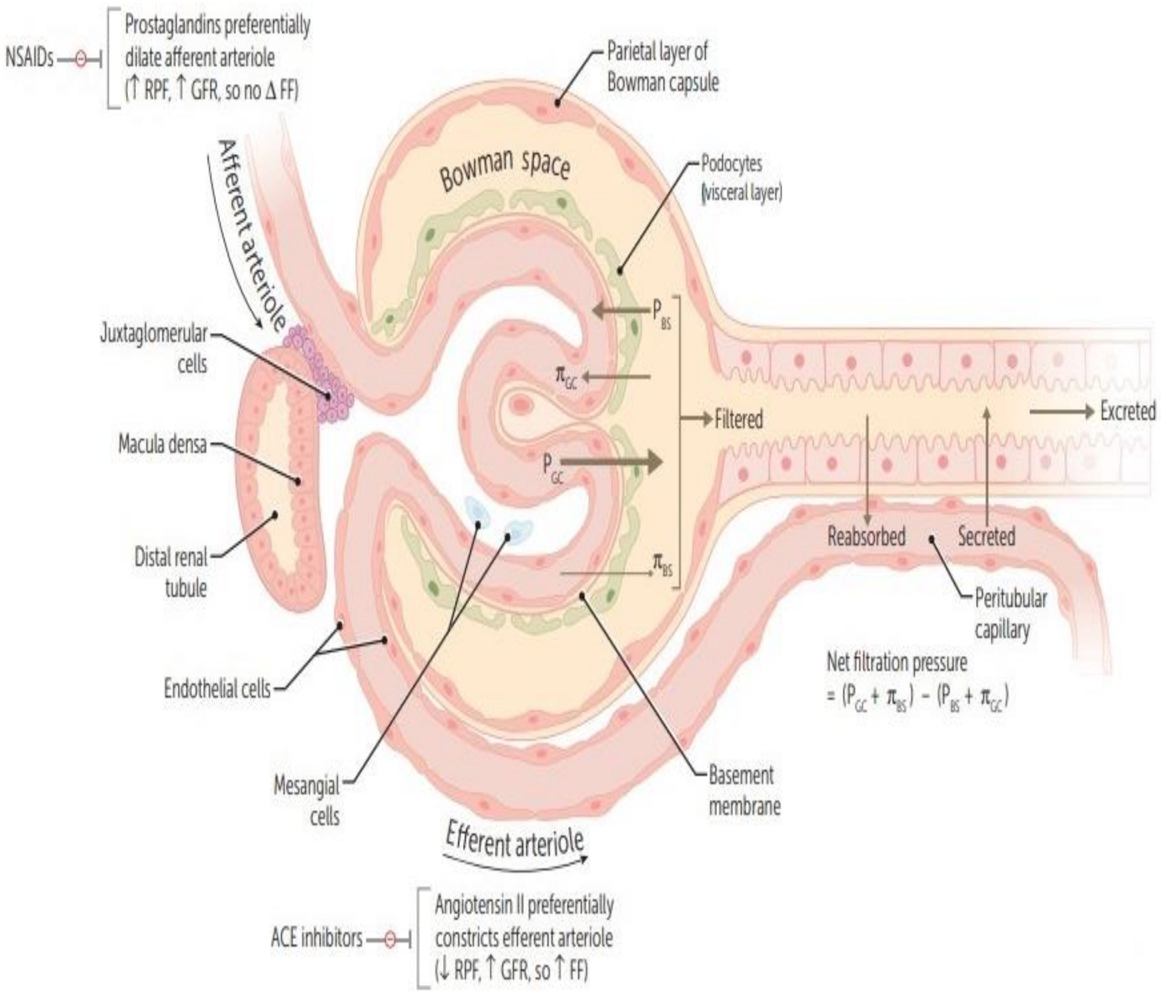
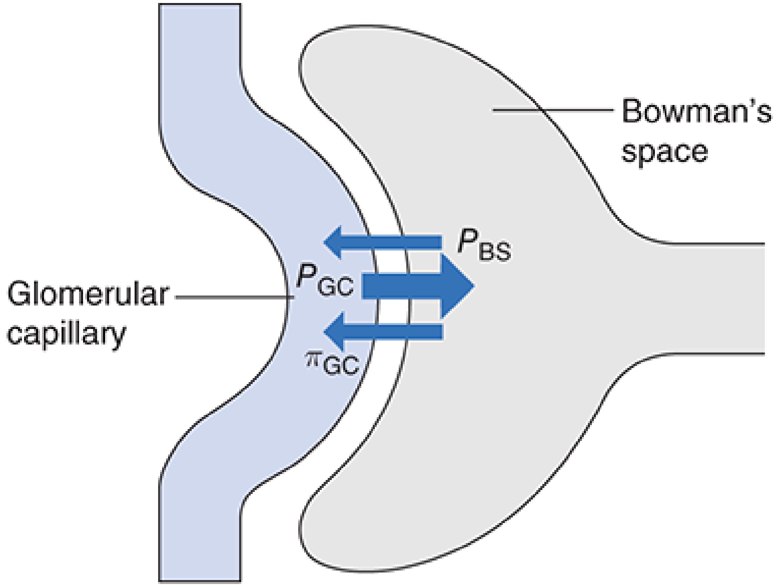




Forces	mmHg
Favoring filtration:	
Glomerular capillary blood pressure (P_{GC})	60
Opposing filtration:	
Fluid pressure in Bowman's space (P_{BS})	15
Osmotic force due to protein in plasma (π_{GC})	29
Net glomerular filtration pressure = $P_{GC} - P_{BS} - \pi_{GC}$	16

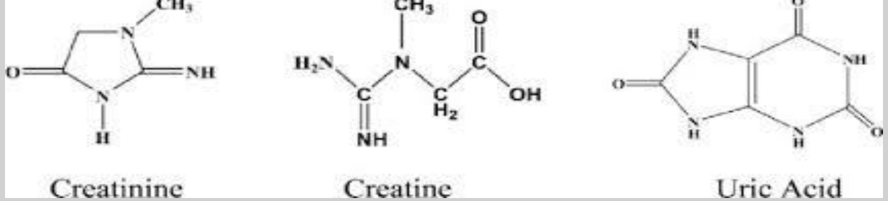
Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

- Le, T., Bhushan, V., Qiu, C., Chalise, A., Kaparaliotis, P., Coleman, C., Kallianos, K., & Evans, S. (Eds.). (2024). *First aid for the USMLE Step 1 2024: A student-to-student guide* (34th ed.). McGraw Hill.

More Filtration Formulas to Know

- **Filtration Fraction (FF)** = GFR/RPF = Clearance of creatinine/clearance of PAH
- **Filtered Load** (mg/min) = GFR*plasma concentration, substance filtered in a given time
- GFR is best estimated with **creatinine** clearance in **urine** (V*U/P)
- eGFR is best estimated with **blood** creatinine
- RPF is best estimated with **PAH** clearance (V*U/P)
- Normal FF is approximately 20%, increasing GFR only increases FF, increasing RPF only decreases FF

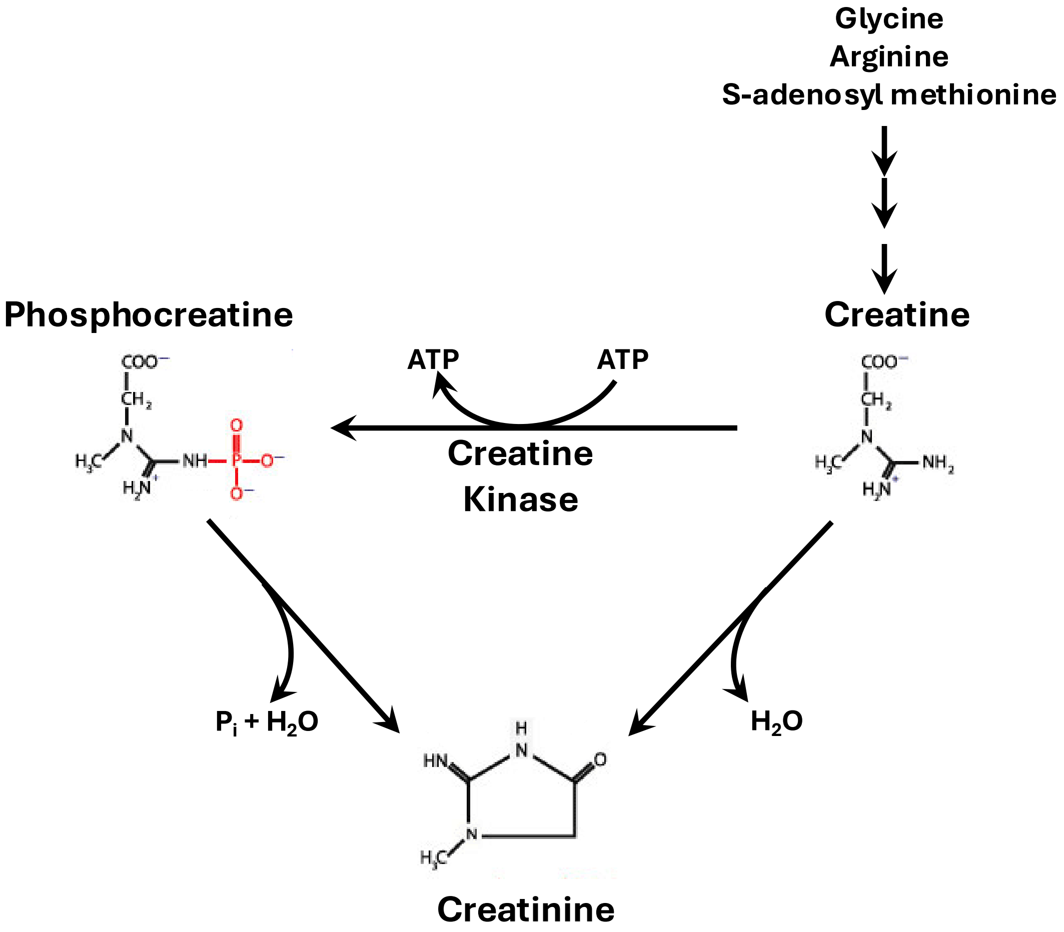
The relationship between **eGFR** and **Clearance**



- **Creatinine** is the closest endogenous marker to **inulin** for estimating GFR, though it slightly overestimates GFR (≈10–20%) due to tubular secretion.
- Normal serum creatinine is ~0.74–1.35 mg/dL in men and ~0.59–1.04 mg/dL in women, but varies with muscle mass, age, diet, and activity. It is a byproduct of muscle metabolism (from **creatine**, an energy storage molecule that can also be influenced by supplementation).
- While **inulin** is considered to be the gold standard (freely filtered, not reabsorbed or secreted), creatinine is more practical for routine use.
- **Cystatin C** is an alternative endogenous marker produced by all nucleated cells and is less influenced by muscle mass or diet, but can be affected by conditions such as thyroid disease, diabetes, and smoking.
- Creatinine and cystatin C are the two most used biomarkers for estimating kidney function (eGFR).

Relationship substance "X"	Rationale	Outcome
Cx<eGFR	eGFR is higher than clearance (Clearance= V*U/P)	Net reabsorption of X or not freely filtered
Cx>eGFR	Clearance is higher then eGFR, similar to creatinine (because of minor kidney tubular secretion)	Net secretion of X
Cx=eGFR	Clearance and eGFR are equal, similar to Inulin	Not secreted or reabsorbed

Creatinine Synthesis

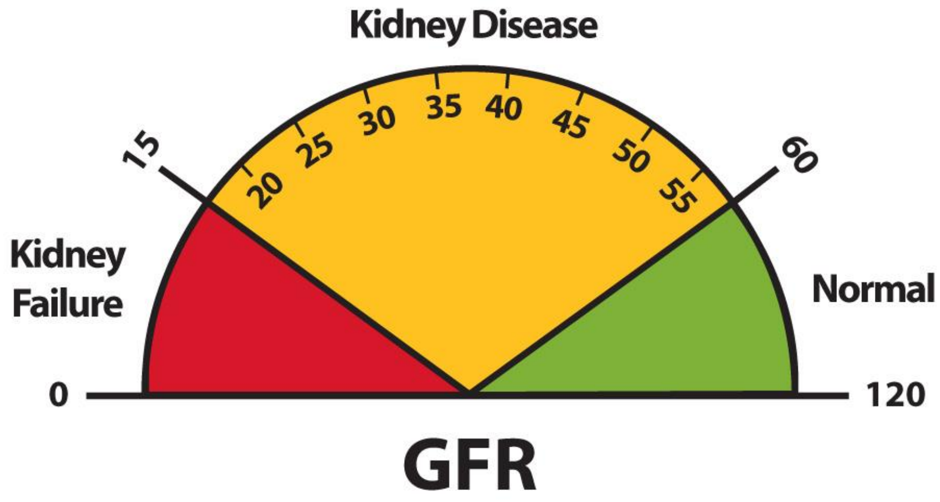


- Present in all body fluids
- Filtered in glomeruli
- Urinary excretion

The importance of eGFR/GFR

https://www.kidney.org/professionals/gfr_calculator

https://www.mdcalc.com/calc/76/mdrd-gfr-equation



The Stages of Chronic Kidney Disease

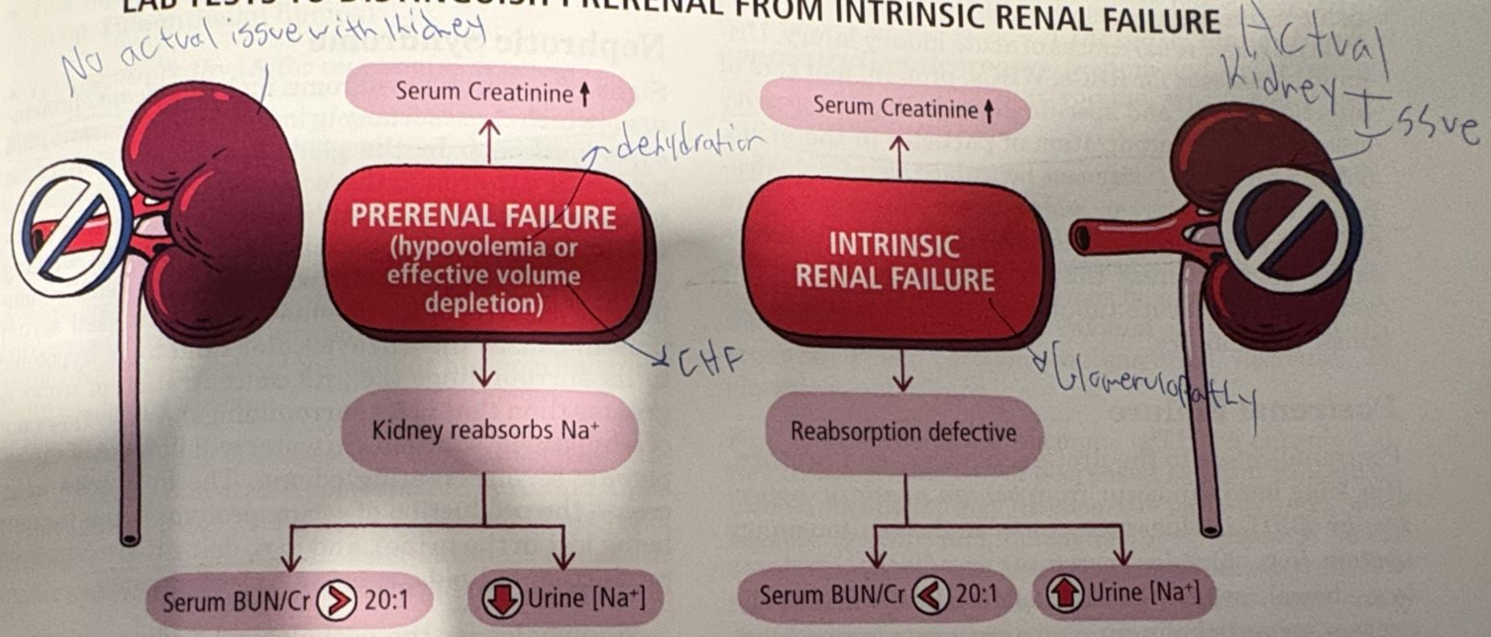
	eGFR	Kidney Damage	Symptoms
Stage 1	90 or above	Mild	Few or none
Stage 2	60–89	Mild	Few or none
Stage 3a	45–59	Mild to moderate	Mild to moderate symptoms: fatigue, swelling, itchy skin, abnormal urine
Stage 3b	30–44	Moderate to severe	Mild to moderate symptoms: fatigue, swelling, itchy skin, abnormal urine
Stage 4	15–29	Severe	Strong symptoms: fatigue, swelling, itchy skin, abnormal urine
Stage 5	Less than 15	Permanent kidney failure	Severe illness; can only be treated with dialysis or kidney transplant

Everyday Health. (n.d.). *Chronic kidney disease (CKD): Symptoms, causes, diagnosis, treatment, and prevention*. Everyday Health. Retrieved June 11, 2026, from https://www.everydayhealth.com/urinary-conditions/chronic-kidney-disease/guide/

ume state (real or effective), and thus actively reabsorb

sic renal failure. In summary, in any type of renal

LAB TESTS TO DISTINGUISH PRERENAL FROM INTRINSIC RENAL FAILURE



	PRERENAL	INTRINSIC RENAL
Serum BUN: Creatinine ratio	≥ 20:1	10-15:1
Urine [Na+]	< 20 meq/L	≥ 40 meq/L
Fractional Excretion of Sodium (FENa)	< 1%	> 2%

Figure 3-2

Berkowitz, A. L. (2007). *Clinical pathophysiology made ridiculously simple* (3rd ed.). MedMaster.

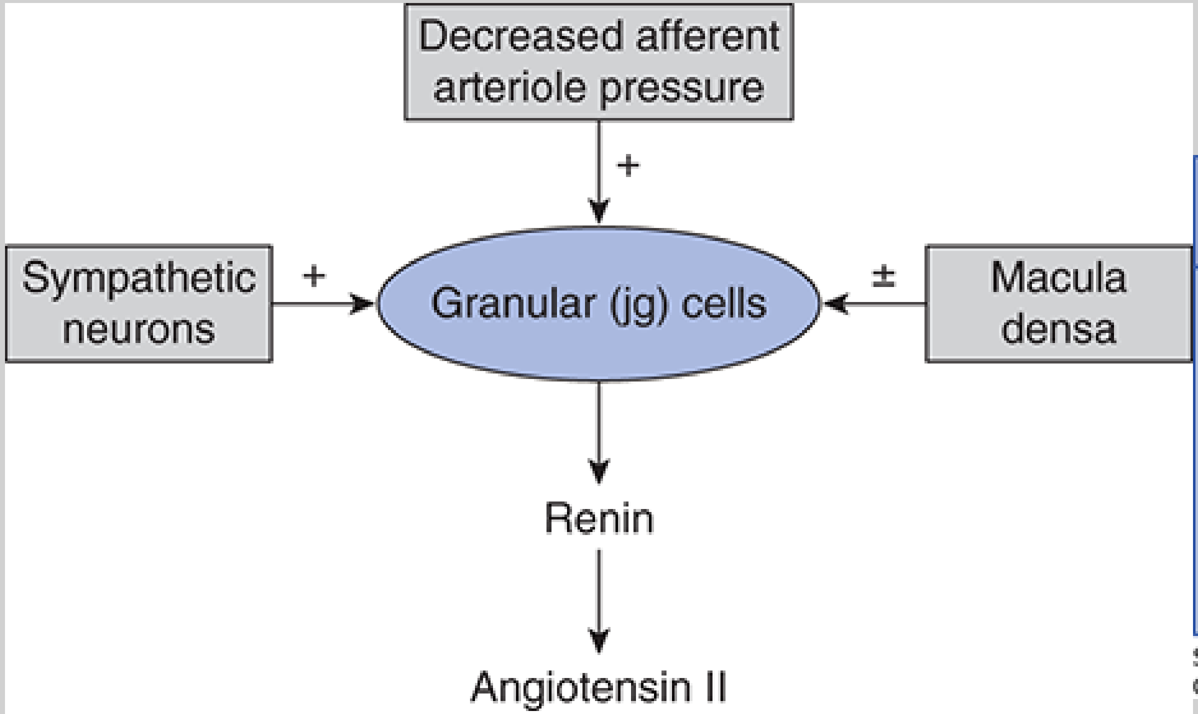
Autoregulation

- The Kidneys want to support constant GFR and RBF despite changes in renal perfusion pressures
- Decreased perfusion due to many mechanism (HFrEF, dehydration, renal artery stenosis, glomerular disease etc...) can lead to damage to the kidneys (AKI), especially within the medulla
- Too much perfusion can overwork the kidneys (such as in diabetes, high salt diets, obesity etc...) leading to eventual collapse of renal function
- Mechanisms including **innate smooth muscle tone, Macula Densa cells(chemoreceptors)/JG cells(mechanoreceptors) and the autonomic nervous system regulate** these changes

1. **Myogenic Mechanism**
    - ↑ BP → afferent arteriole stretch → vasoconstriction → ↓ RBF + ↓ GFR (proportional; FF unchanged)
    - ↓ BP → ↓ stretch → vasodilation → ↑ RBF + ↑ GFR

2. Tubuloglomerular Feedback Mechanism

- ↑ NaCl at macula densa → ↑ GFR signal → ATP + thromboxane A2 → afferent constriction → ↓ GFR + ↓ RBF
- ↓ NaCl at macula densa → ↓ GFR signal → prostaglandins + renin (RAAS) → afferent dilation → ↑ GFR + ↑ RBF



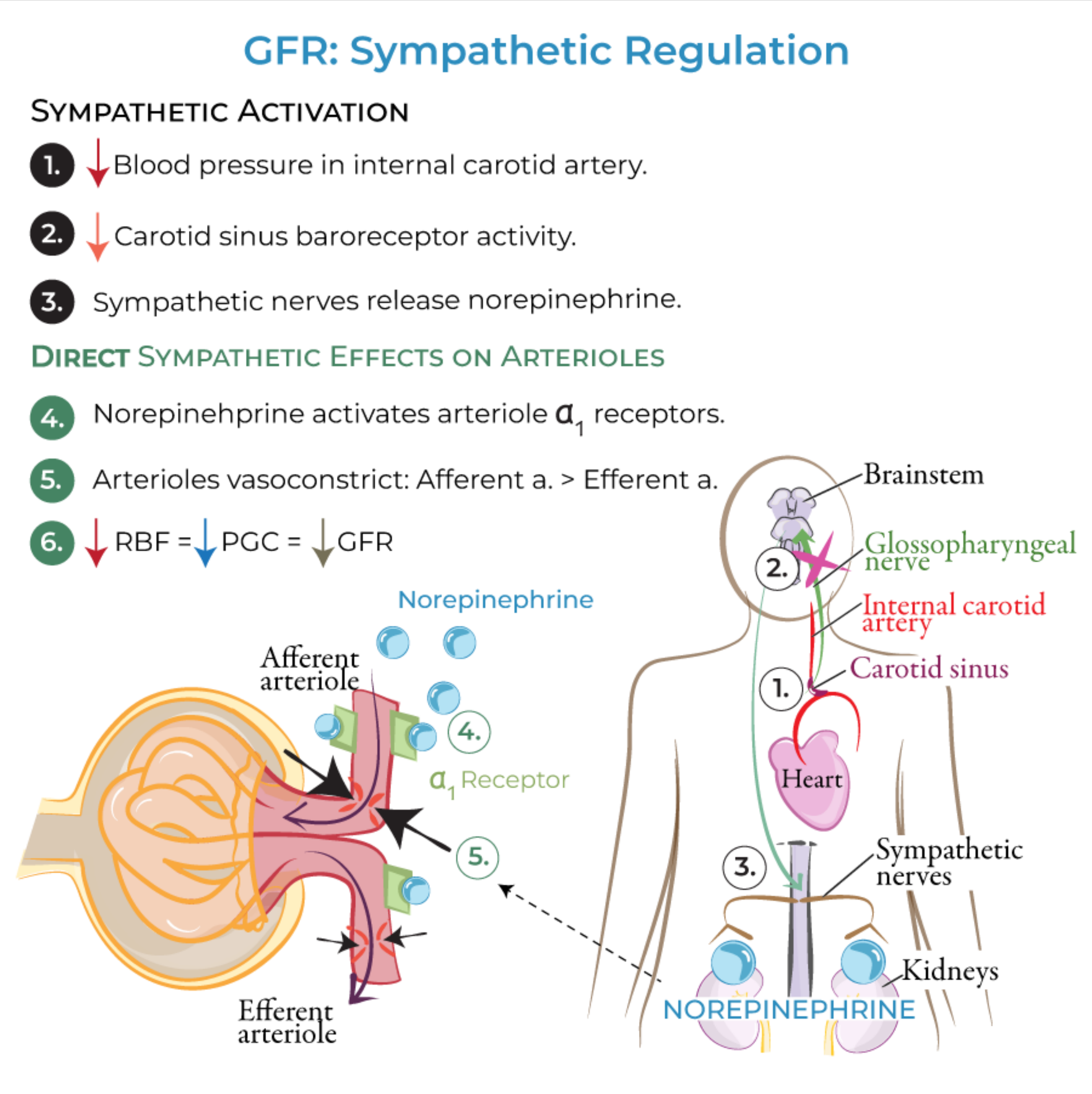
Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

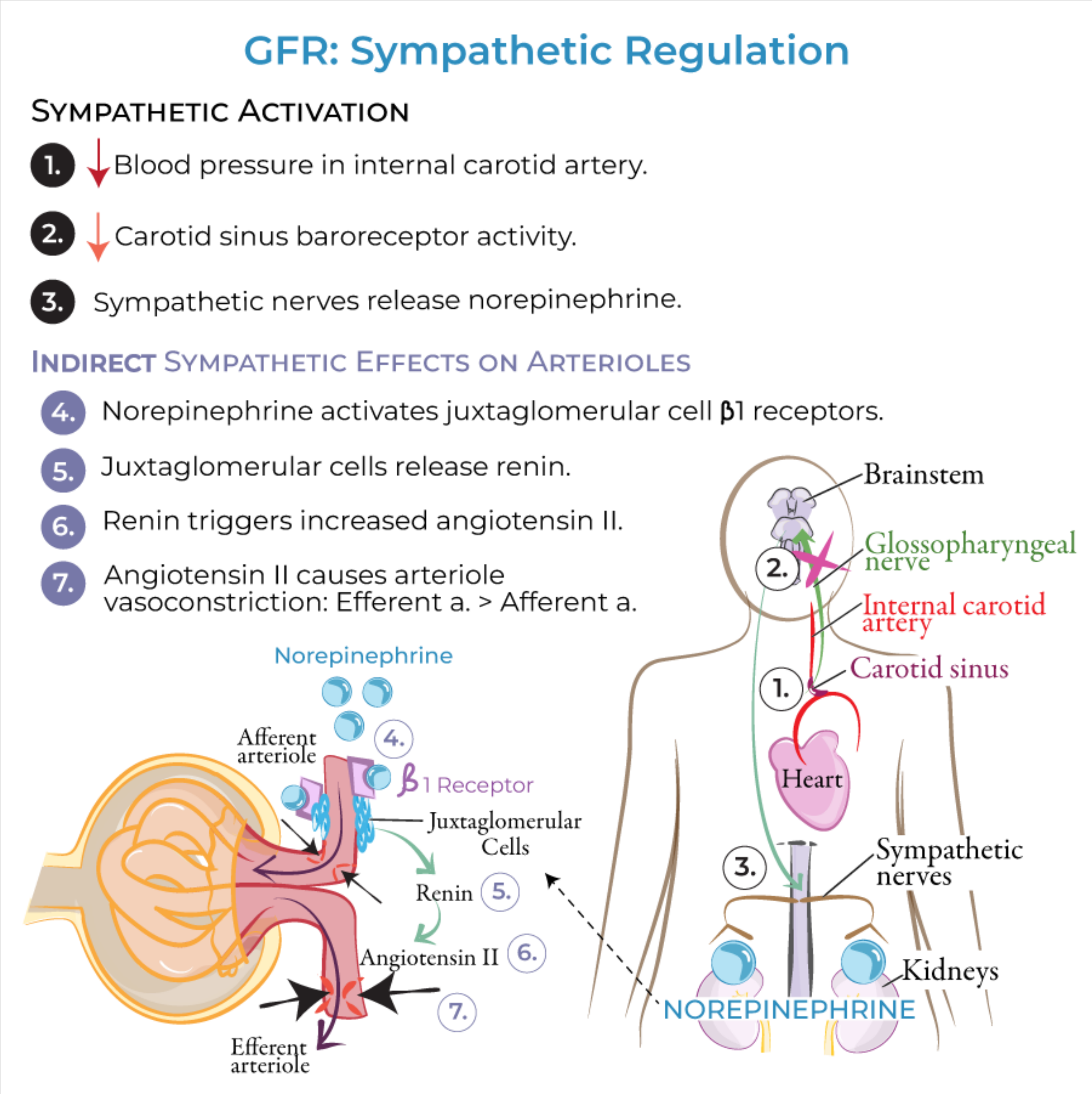
Condition sensed by macula densa	Mediators released by macula densa	Actions
↑Na delivery	ATP (adenosine)	↓ GFR, ↓ renin secretion
↑ Tubular flow	NO	Blunts reduction in GFR
↓Na delivery	prostaglandins	↑ Renin secretion

Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

T10-T11 Sympathetic Kidney/Adrenal Levels. **Vagus Nerve (CN X)**- Parasympathetic.

Anteior Chapman- Kidney- 1-inch lateral, 1 inch superior to umbilicus. Adrenals- 1-inch lateral, 2 inches superior to umbilicus





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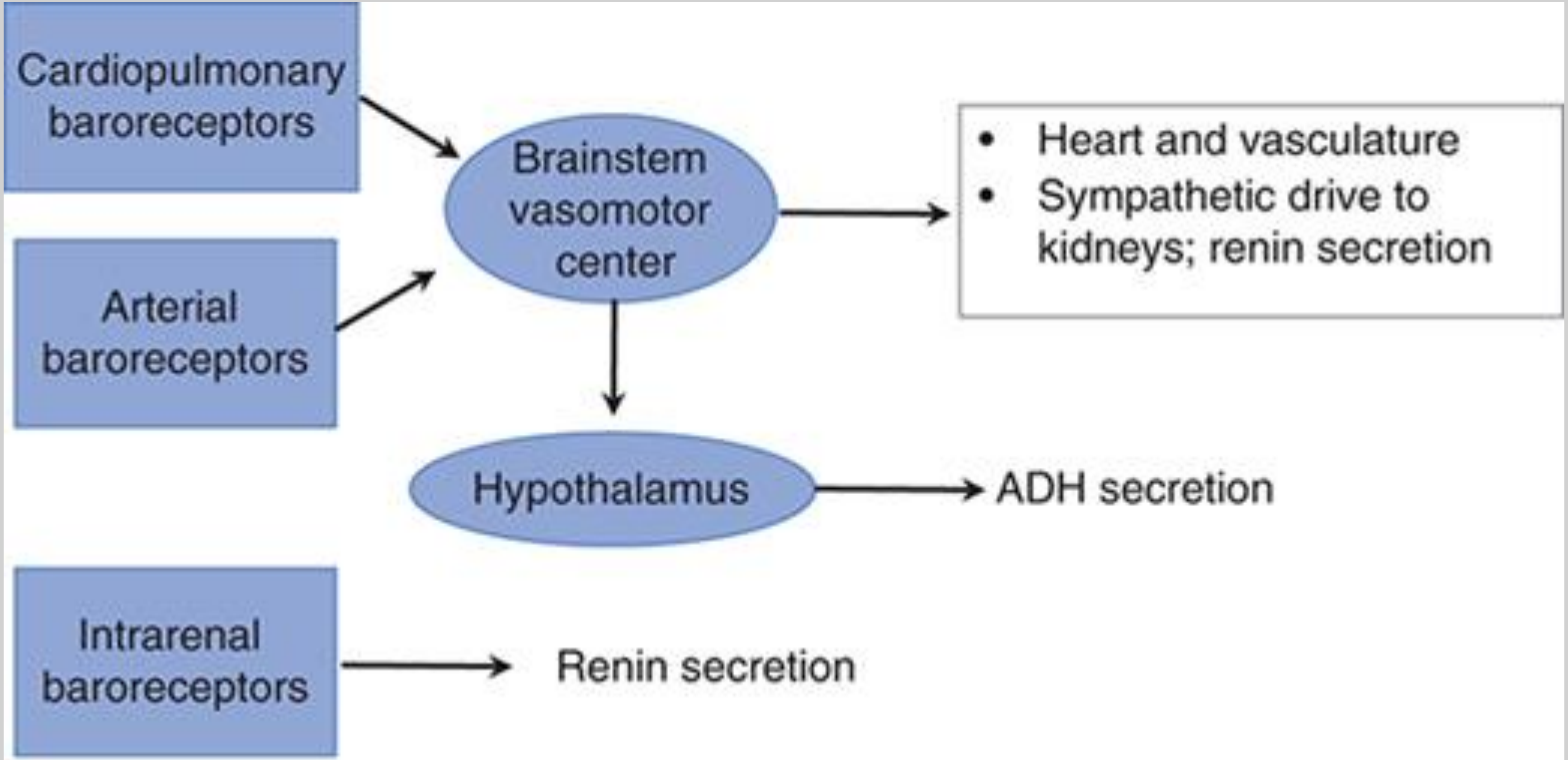


Source: Douglas C. Eaton, John P. Pooler: *Vander's Renal Physiology, 10e*

Vasoactive Substance Influence

- **Vasoconstrictors**
  - **Renin**- RAAS system initiating enzyme, made by JG cells in the kidney, catalyst for angiotensinogen (made in liver) to angiotensin 1
  - **Angiotensin 2 (Ang 2)**- RAAS system end-product hormone, converted from angiotensin 1 by ACE (made in lung), affects across the body, preferentially constricts the efferent arteriole in the kidneys (greater concentration of AT1 receptors)
  - **Norepinephrine (NE)**- Produced in the Locus coeruleus (brain), sympathetic ganglia, and binds to Alpha-1 receptors (G_q)
  - **Endothelin (Peptide)**- Produced by kidney endothelial & tubular cells, paracrine/autocrine, imbalance occurs in various pathologies
  - **Anti-Diuretic Hormone/Vasopressin (ADH)**- Produced by hypothalamic neurons, stored & secreted by the posterior pituitary (neurohypophysis), acts on the collecting duct V2 receptor, leading to an increase in aquaporins (AQP-2) and increased water reabsorption
  - **ATP**- Energy molecule that extracellularly acts on purinergic receptors in the kidney, released by Macula densa cells
  - **Thromboxane A_2 (TxA_2)**- Lipid mediator derived from COX enzymes/arachidonic acid that is involved in platelet aggregation, increased release in inflammatory conditions, made by platelets, macula densa cells, podocytes, and endothelial cells, G_q mechanism
  - **High dose Dopamine**- D1 and D2 receptor activation as well as α_1 adrenergic receptors (G_q; Ca^{++} influx)
  - **Cocaine and Methamphetamine**- Block NE and dopamine re-uptake, increasing α-1 (G_q; stimulation, increase endothelin and decrease nitric oxide

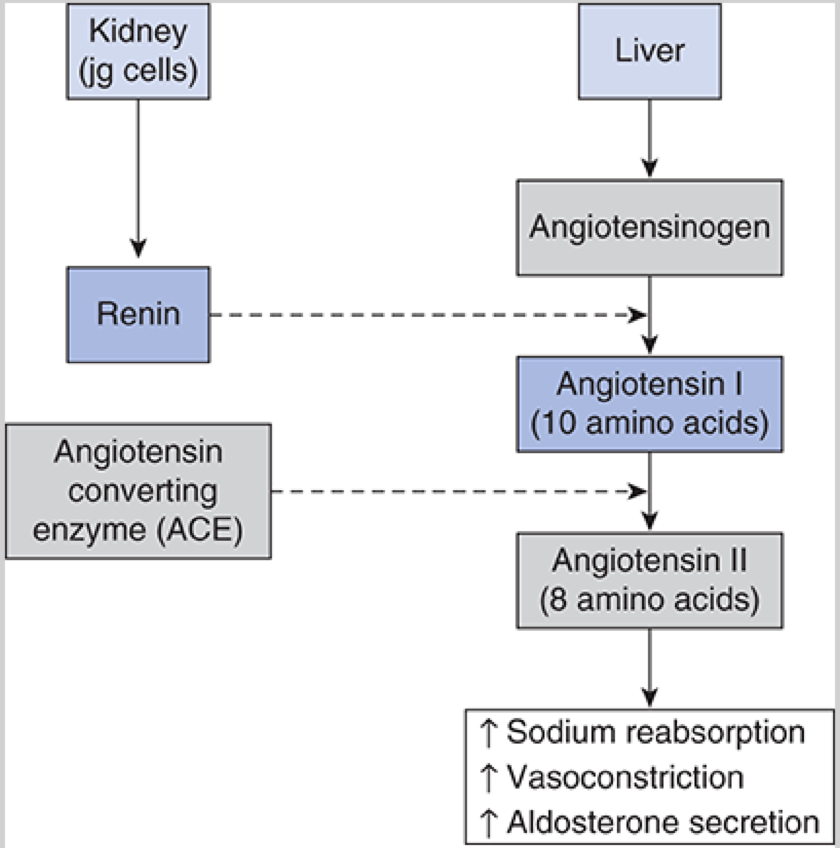
- **Vasodilators**
  - **Prostaglandins (PGE_2/PGI_2)-** Synthesized locally by kidney cells COX-1 and COX-2 enzymes, increase cAMP, during stress and volume depletion save kidney from ischemia (act as a "shield"), relax mesangial cells and afferent arteriole preferentially
  - **Nitric Oxide-** Keeps a basal level of relaxation in kidney vessels, Comes from amino acid L-arginine, diffusible gas across cells, decreases calcium release, and increases smooth muscle relaxation, without it there would be immense renal vasoconstriction
  - **Atrial Natriuretic Peptide (ANP/BNP)-** Made by atrial myocytes within the heart, decreases total body salt and water
  - **Kallikrein/Bradykinin-** Produced in the distal nephron, leads to Prostaglandin and NO local release
  - **<u>Low dose</u> Dopamine-** Acts on D1 and D2 receptors (cAMP pathway leading to PKA and vasodilation) **without** interacting with Alpha-1 receptors

Think about each of these substances and how increasing them or decreasing them would affect kidney physiology

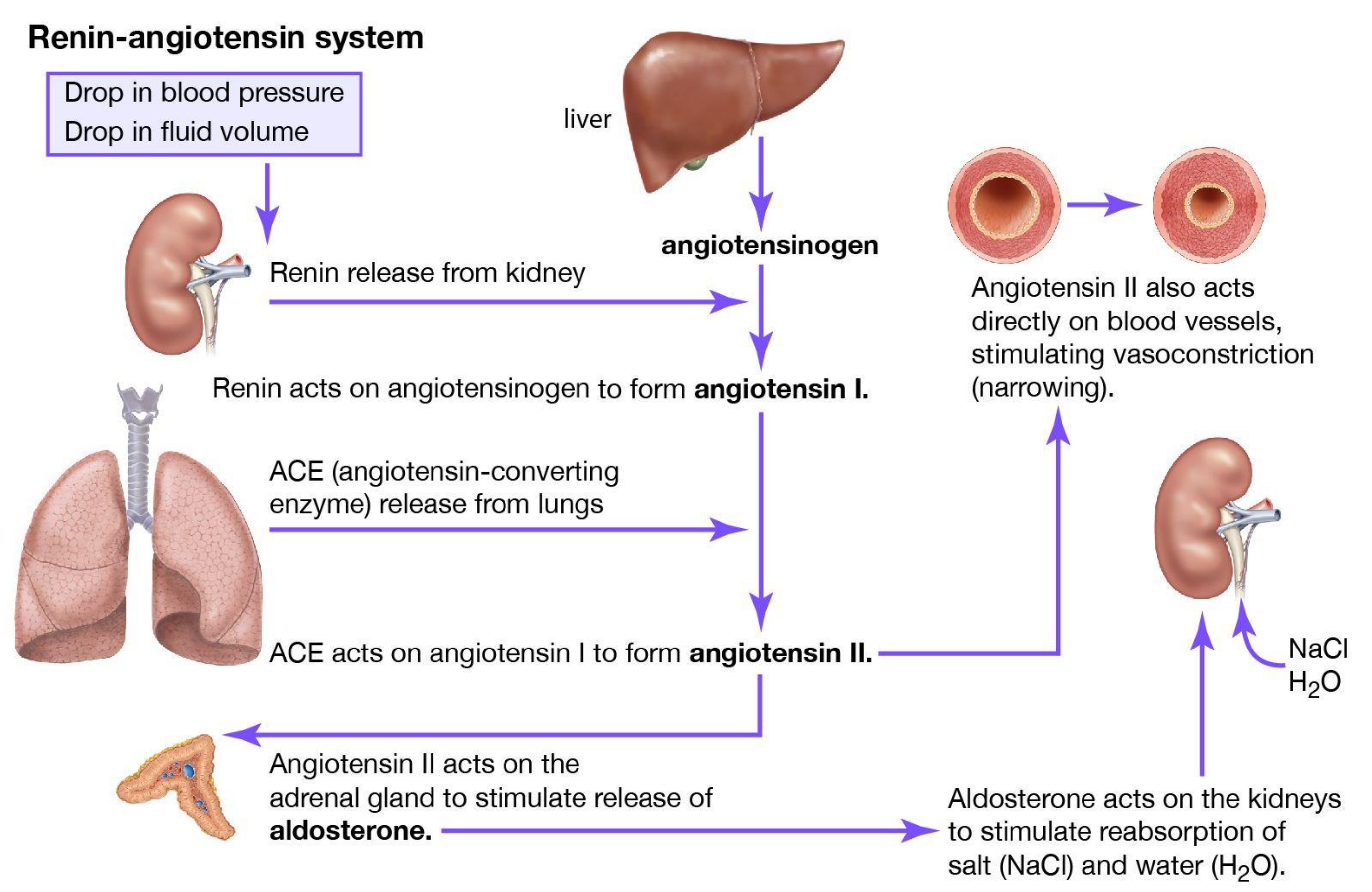
For example, what effects would something that inhibits ACE (Lisinopril) have on GFR, Blood pressure and the RAAS system?

How would something that decreases prostaglandins (NSAIDs) effect tone of afferent and efferent arterioles?

What would blocking V2 receptors do?



Source: Douglas C. Eaton, John P. Pooler:
Vander's Renal Physiology, 10e



The RAAS System (Covered in depth within other lectures)

Pathologies related to Filtration and Blood Flow

- Glomerulonephritis/Nephritic Syndrome
- Heart Failure
- Liver Cirrhosis
- Multiple Myeloma
- Amyloidosis
- Chronic Kidney Disease
- Dehydration and Hemorrhage
- Diabetic Nephropathy
- Nephrotic Syndromes
- Infectious disease- Pyelonephritis, TB, Malaria, Leptospirosis etc...
- Chronic Hypertension
- Acute Kidney Injury
- Goodpasture
- Rhabdomyolysis
- Nephrotoxic Drugs- NSAIDS, Lithium, Amphotericin B, Acyclovir, Cisplatin, Allopurinol etc..

Back to the case....

Vander's Textbook Questions

- 2–1. *Blood enters the renal medulla immediately after passing through which vessels?*
  - *Arcuate arteries.*
  - *Peritubular capillaries.*
  - *Afferent arterioles.*
  - *Efferent arterioles.*

2–2. *Which cell type is the main determinant of the filterability of plasma solutes?*
  - *Mesangial cell.*
  - *Podocyte.*
  - *Endothelial cell.*
  - *Vascular smooth muscle.*

- 2–3. *Which one of the following is NOT subject to physiological control on a moment-to-moment basis?*
  - *Hydrostatic pressure in glomerular capillaries.*
  - *Selectivity of the filtration barrier.*
  - *Renal blood flow.*
  - *Resistance of efferent arterioles.*
- 2–4. *A substance is freely filtered and has a certain concentration in peripheral plasma. You would expect the substance to have virtually the same concentration in*
  - *the glomerular filtrate.*
  - *the afferent arteriole.*
  - *the efferent arteriole.*
  - *All of these places.*
- 2–5. *In the face of a 20% decrease in arterial pressure, GFR decreases by only 2%. What could account for this finding?*
  - *The resistances of the afferent and efferent arterioles both decrease equally.*
  - *Glomerular mesangial cells contract.*
  - *Efferent arteriolar resistance increases.*
  - *Afferent arteriolar resistance increases.*

- Taken from: Eaton, D. C., & Pooler, J. (2023). *Vander's renal physiology* (10th ed.). McGraw Hill. Chapter 2 end questions

- 2–6. *The hydrostatic pressure within the glomerular capillaries*
    - *is much higher than in most peripheral capillaries.*
    - *is about the same as in the peritubular capillaries.*
    - *decreases markedly along the length of the capillaries.*
    - *is generally lower than the oncotic pressure in the glomerular capillaries.*
- 2–7. *Autoregulation serves to maintain near constancy of*
    - *the amount of filtered solute.*
    - *glomerular capillary pressure.*
    - *the amount of filtered water.*
    - *All of these.*
- 2–8. *The glomerular filtration barrier is*
    - *more permeable to small cations (e.g, sodium) than small anions (e.g, chloride).*
    - *freely permeable to small peptide hormones.*
    - *freely permeable to all uncharged molecules.*
    - *freely permeable to all substances in the blood except blood cells and platelets.*
- 2–9. *The term filtered load is*
    - *another name for GFR.*
    - *the amount of a substance entering the kidneys in the renal arteries.*
    - *the amount of a substance filtered in a given time.*
    - *the concentration of a substance in the glomerular filtrate.*
- 2–10. *Oncotic pressure in renal blood*
    - *is much higher in blood entering the glomerular capillaries than in blood entering peritubular capillaries.*
    - *rises during passage through the glomerular capillaries.*
    - *is constant throughout the renal vasculature.*
    - *is higher in afferent arterioles than in efferent arterioles.*

- Taken from: Eaton, D. C., & Pooler, J. (2023). *Vander's renal physiology* (10th ed.). McGraw Hill. Chapter 2 end questions

Board Style Questions

1. In a novel drug clinical trial, a healthy 24-year-old male volunteer receives an infusion of a drug in the NSAID class that has not been FDA approved for release. This drug has been previously proven to have a similar MOA to ibuprofen. The purpose of this trial is to see how the drug can affect renal physiology. The participant is 5'10, 195 pounds, current HR is 75 bpm, and blood pressure is 132/85 mmHg. All other vitals are within normal limits (WNL). He takes no medications and reports NKDA. He is in good health with no medical or psychiatric conditions and admits to having no past surgical history except an uncomplicated adenoidectomy at 12-years old. He currently attends graduate school, sleeps 6-7 hours a night and has no family history of kidney disease. Based on knowledge of how typical NSAIDs effect kidney physiology, which of the following changes is most likely to occur in this participant's kidneys after administration of this new medication?

Renal Blood Flow	**Glomerular Capillary Hydrostatic Pressure**	**GFR**
A.	↑	↑
B.	↑	↓
C.	↓	↓
D.	↓	↑
E.	No change	No change

2. A healthy 28-year-old man with a previous history of GERD and GAD is found to have a plasma oncotic pressure of 32 mmHg, a glomerular capillary hydrostatic pressure of 60 mmHg, a Bowman's space hydrostatic pressure of 18 mmHg, and a Bowman's space oncotic pressure of 0 mmHg. What is the net filtration pressure driving GFR?

A. 10 mmHg

B.14 mmHg

C. 42 mmHg

D. 28 mmHg

E. 60 mmHg

3. A 34-year-old female presents to the ED with recurrent panic attacks. She states that early that day, she had one attack so intense she couldn't leave her house to get to work. Despite lasting 10 minutes, it was so exhausting and debilitating that she felt she needed to get emergency care before trying to return to work the following day. At the time of arrival, her vitals are within normal limits (WTN); she is alert and oriented X 3 and shows no signs of distress but some fatigue. Suddenly, while being interviewed by a resident physician, she starts to sweat, her pupils dilate, and stops answering the residents' questions. She starts to visibly show discomfort and appears in acute distress. Her mean arterial pressure suddenly rises from 90 mm Hg to 130 mm Hg. Despite the increase in MAP, her kidneys keep her GFR relatively stable. Which homeostatic mechanism is primarily responsible for keeping her GFR constant?

A. Increased aldosterone secretion

B. Activation of the renin-angiotensin system

C. Tubuloglomerular feedback and myogenic response

D. Increased sympathetic stimulation

E. Increased ADH release

4. While studying snake renal physiology, you discover a new vasoactive agent which does not exist in mammals. This molecule is a peptide hormone, and you decide to name it "Slither 1". It is figured out that "Slither 1" among other things, preferentially constricts the efferent arteriole. Which hormone acts an analogous way in humans and what hemodynamic changes would be expected at low-to-moderate "Slither 1" levels in the snake's kidney (assuming snake physiology is identical to human)?

A. Angiotensin II, decreased glomerular capillary hydrostatic pressure and decreased GFR

B. Angiotensin II, increased glomerular capillary hydrostatic pressure and increased GFR

C. Angiotensin I, increased renal blood flow with no change in GFR

D. ANP, decreased filtration fraction with increased plasma flow

E. Angiotensin II, decreased oncotic pressure in glomerular capillaries

5. A 32-year-old female ice hockey goalie was playing in WHL finals when an opposing player accidentally tripped, colliding into her and slicing her jugular vein with her skate blade. An EMT and paramedic bystander in the crowd rushed to her aid but could only compress hard to enough to get her in the ambulance and rush her to the nearby hospital. At the hospital, her pulse is thready, but around 150 bpm, BP is 82/59 mmHG, RR is 38 breaths/min. She is cold and clammy, confused and lethargic. As she's being treated, she is also found to have acute oliguric renal failure, with a non-physiological drop in GFR. Which combination of vasoactive effects best explains the drop in GFR?

A. Prostaglandin E_2 dilation of afferent arteriole and NO release

B. Sympathetic-mediated afferent arteriolar constriction and high angiotensin II levels causing efferent constriction that overcomes glomerular pressure

C. CANP-mediated efferent constriction and afferent dilation

D. Dopamine-mediated vasoconstriction of both arterioles equally

E. Endothelin dilation of the efferent arteriole

6. A 24-year-old man presents with hemoptysis, fatigue, and dark-colored urine for 2 weeks. Physical examination is notable for mild hypertension. Laboratory studies show elevated serum creatinine and numerous red blood cell casts on urinalysis. A kidney biopsy proves crescent formation in Bowman's space. Immunofluorescence reveals a linear pattern of IgG deposition along the glomerular basement membrane.

The patient's renal disease is caused by autoantibodies directed against a major part of which of the following structures involved in the glomerular filtration barrier?

A. Fenestrated endothelial cells of the glomerular capillaries

B. Type IV collagen within the glomerular basement membrane

C. Podocyte foot processes forming the slit diaphragm

D. Mesangial cells supporting the glomerular capillaries

E. Proximal tubular epithelial cells responsible for protein reabsorption

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